

ActInf Livestream #021.03 ~ John Boik

Livestream | John Boik | 2:08:53

okay

hello and welcome to active inference

lab this

is the third in a ongoing series of

discussions

between i guess active inference lab and

john boyk

i'm daniel i'm a postdoc in california

and i'm going to be kind of

working through the jam board as we've

done in these other

sessions while john goes through his

paper

some of the implications we can also

raise up some questions from the live

chat so people are totally

welcome to um leave questions in there

wait actually it's already sort of

messed up

or not sure i hope that it will work it

looks like it's okay

we're just sort of dealing with the

streaming tech and i'll play with it

once you start talking jon but um yep

i hope this uh is a fun discussion

because the other ones have been awesome

as well

so i'll flip over to slide 14 from our

second

slideshow and let you pick up with some

points from last time

and uh let you know if anything is awry

okay sounds good thank you my name is

john boyk i'm

a courtesy faculty at oregon state

university environmental science

graduate program

and this series is about our series of

discussions is about a

series of three papers that i recently

published in the journal sustainability

and the topic is societal transformation

um i think somewhere in the middle of

this talk we'll show some links to those

papers and

and to my website where all this

information can be found

uh so we're do you have the slides up

you want to put up slide 14

we're just going to jump in we're going

to jump into halfway through the last

talk we had a bunch of slides that we

didn't get to

so we're going to try to get to them

um now yep i'm there so i'll be

following along but you just won't see

it but we're good to go

oh i won't see it okay all right okay

all right so uh slide 14

is uh is a list of 14 desired data that

i had

proposed in this second the second paper

part two

uh i should probably say what the title

of that series is the title is science

driven societal transformation parts one

two and three

so we were talking about part two last

time which

was on um

motivation and strategy and we're going

to continue with that now

um so in that part two paper i had

listed 14 desert data for what any
what would be what maybe could or should
be in any r
d program aimed at societal
transformation
um throughout the series i touch on all
of these fourteen
um i would just like to make
clear that what i'm proposing the series
the three-part series is itself
a proposal for an r d program it's kind
of an informal proposal to the world
about
a program that could be started and
it is possible that other people have
other groups have other ideas of what
kind of r d
program you know could be constructed
and
you know it could be that in the future
there's
numerous uh r d programs that are in
operation that are all looking at this
topic of societal transformation
and i wanted to offer a list of uh of
points or topics that that each of them
might
address and that way at the very least
we could compare programs by comparing
these for
14 points
um i just we we covered we went through
him last time so i'm not going to go
through him again but i just wanted to
say a couple of things
uh daniel i remember from our last
conversation you had said something
about

um you know getting a a bunch of
scientists together and maybe they won't
agree on
you know different aspects of this and
there you know could be
some arguments for
one point over another point or one
approach over another approach
and that of course that is true and
that's fine and that's good there
should be a variety of voices uh heard
and considered in in any r d program
but i i did want to make the point that
the r d program that i'm proposing the
the concept
is evidence-based science-driven r d
program
so um suppose
that uh well let me say that
that an evident evidence-based program
is different than a popularity-based
program
so it's not just that you know everyone
gets to vote for what they want and
it'll
and and the r d program would go off in
in whatever direction
everyone votes for i mean in in a sense
that's true because
how else would you govern things but in
another sense it's not true because
there has to be some kind of
science process i mean we have to be
addressing with the
with a design program we have to be
addressing the
the science-based aspects of
it so um and that is largely driven by

fitness

purpose and fitness so the so

all right um let's see

yep just continue going ahead thank you

okay we're good yep

okay all right uh i was saying that um

that there could be multiple p multiple

uh

uh r d programs proposed by different

groups

driven by different groups but i would i

would expect that they would all be

based on some kind of science-based

evidence

approach so while while each

r d program might have a slightly

different way of measuring fitness

those discussions would be based on

scientific evidence and reasonable

you know reasonable evidence as opposed

to just a popularity

we're good yep working with tech

thank you everyone for standing by when

it kind of flips out for a second

we'll try to restart it like as soon as

possible but that's why we use the

asynchronous

modes of work because we're really lucky

when we get to have synchronous time

with the authors or with anyone but the

asynchronous

and the updates on the comments are what

really

matter for this kind of a project not

someone's internal

understanding or just a dialogue so

thanks for bearing with us everyone go

ahead john

okay just to make that last point is the
r d program that i've proposed
is really it's really a meta meta
framework for learning new systems
new system designs so the idea is that
the
the r d program would look at a wide
variety of
of designs people people it would
provide a way
a means for evaluation of different
designs that different groups might come
up with
so that's all i wanted to cover on that
topic and then on to new information
so

okay

yep okay we'll be able to do it right

okay here is this

this is going to work give me one second

all right thanks everyone for holding by

i hope that you

enjoyed this section

the silence to think about these ideas

okay so now

it's called making lemonade out of yeah

okay

so now i will spend some of the time

when you're talking

john to just continue like trimming and

stuff but

we actually um are back

in the game so please continue

on 5 15 and 2 and go from there and then

continue on so sorry about that first 15

minutes of fun

now go for it just one little thing my

screen

my picture on my computer is quite dark
but i don't know if i'm coming through
on everyone else's computer
looks good and i'm going to fix it on
the other computer okay all right
all right so slide 15. uh
uh i was saying before before the
interruption
that uh this r d just a few things on
the rd program it's a
it has a as i envisioned at a 50-year
horizon so this
is um this is a careful movement
towards societal transformation it's
going to take a
a a long time but um
there will studies will be if it were
funded if it
moved forward studies would start right
away field trials would
follow in in the near term and those
communities that
choose to participate in field trials
would see you know
benefits far faster than than than
in other communities that are waiting so
this is a 50-year horizon towards
a global scale or no near global scale
transformation so it's going to take a
while
to to achieve change at uh you know at
that
huge scale um
uh i also want to mention that the
program involves the public in a variety
of ways this is i
envision this as a partnership between
the public

especially between local communities
and the science the global science
community
so there would be field trials there
would be a variety of studies and the
public can get involved in a variety of
ways
there would be lab lab studies there
would be surveys there would be
the ability for the public to in
input their ideas and their their wishes
and desires into the program so
hopefully this would be a
a conversation a multi-way conversation
between the stakeholders
that are involved and uh on the science
side
uh i mostly want to
make the point that the program could
involve nearly every academic field
that's out there literally in a to z
and at the bottom of this slide i just
list a few of the many
fields that would be appropriate here
[Applause]
certainly some of the the science fields
have been talking about complex science
and
other fields like that but also
education
media literature linguistics
like you name it and there would be a
place a need for
participation in this project
okay next slide 16.
okay we're good yep all right
uh so um
you know the idea that the science world

would be
actively participating in a program for
societal transformation this is
you know in a sense this is rather bold
and new
uh novel um there there obviously are
programs aimed at the study of of
transitions and
and and
developing the knowledge base for
societal trans uh
some kind of a societal transformation
because many
in the science world and the public
policy world understand that bold change
is necessary if we're going to uh you
know uh
avert the avert problems that we're
facing
due to climate change uh bio
biodiversity loss and a number of other
very serious
social and environmental problems so big
change
deep change is on many many uh
people's minds but
uh science uh the role that science
has that many people envision for
science is more of a knowledge
is more than the traditional science
world the knowledge
gathering knowledge product production
that would support the public and and
politicians and choosing new policies
and various things
so the idea that science would be
involved in the actual
r d of making new systems designing new

systems testing new systems
that's more of a what is
being called second order science and
that is fairly new in the science world
so this is uh this is breaking new
ground
for uh force for the science world
so i i have a bullet list here of
various reasons why
the science community might want to be
involved in this
so i'll just go through a few of these
real quick um
first societal the idea that society is
a
can be understood as a organism a super
organism if you will
and a cognitive organism at that is
is timely and important and
offers numerous opportunities to advance
basic
science knowledge so um so
while in in participating in a program
of
societal transformation a great amount
of
science knowledge would be generated uh
next is systems are societal systems are
sophisticated
so uh the the skills that are offered by
the science community are extremely
important
uh they would have to be computers
computer simulations for example and a
variety of other very technical
uh problems that would have to be solved
in order for this for
new systems to really work and for the

design process to work
so uh for that you know the science
community is uniquely
uniquely qualified to contribute to
those effects
uh the goals of transformation are
ambitious
this is the third bullet and yet the
time available for action is short so
this has to be
not tinkering but moving in a direct and
focused way towards societal
transformation and the developing new
systems
and the science community can help with
that to keep it all on track and moving
forward
i've already mentioned the second order
science threshold
threshold to a new way of doing science
and that can be useful
and involve science more in the public
sphere in the future
and the next is that the science
community is trusted by the global
public so
uh and is a bishop and then is a
position to help focus public attention
on
pressing problems and and solutions
next to the last one is the science
community can offer a global
perspective and global coordination so
what we would not want
are a whole bunch of experiments to
occur for the r d program and then to be
forgotten and never picked up again or
never integrated into the body of

knowledge after that
so the science community can offer a
central
in a sense a central focal point a
repository for data and for assessing
data and for assessing the many
experiments
that might occur in such a r d program
and lastly the science community can act
as a powerful ally to the public
uh in promoting and ensuring that
transformation moves forward
so i hope that i hope that the science
community will find interest
in this kind of r d project that's
really the purpose of the whole series
is to
is to offer the the idea to the science
community that
we could do this together this could be
a very important and timely
uh service that the science community
offers to the world
all right on to 17
all right so
and i said this before but if i were
the smartest person on the planet and i
came up with a fantastic
new design for societal systems and and
for viewers that are new by societal
systems i mean economic systems
financial systems
governance systems educational systems
legal systems and the like
um if i were the smartest person i came
up with
perfectly designed societal systems
if there was no way to implement them

practically then my effort would have
been waste wasted
so any r d program aimed at societal
transformation has to have
some kind of viable strategy for
implementing any new designs that might
be created
and uh in this in the proposal in the
series i i
i offer a theory of change and a
strategy for change that i call
local global viral and the idea
there is the local global is the idea
that local
that much of the work would occur at in
local communities
and by local communities i mean
volunteer communities
volunteer individuals who would develop
who would
uh participate in the in the in the
testing of new systems
and and even the design of new systems
uh in cooperation with and partnership
with the global science community so we
have the local global
connection there and
and the viral connection is such that
the
the concept of the of the
i i use what i call a club model and the
concept of the clubs is that clubs would
be implemented locally but on a
volunteer basis as i mentioned and they
would be designed
such to produce substantial benefits for
the participants
and if that is if they're well designed

and indeed the participants
obtained substantial benefits then the
concept of the the technology and the
concept for the clubs and the
club model and the and the systems that
they embody
would quite naturally spread from one
community to the next uh
if if a if a club starts in a
in say in a city just a not not as a
city but just even a small
fraction of a city that that could be a
club any
like a club could be any uh maybe
perhaps a thousand people or more so
just a subset of a city
could start a club and be part of this r
d pr
or the whole r d program a field trial
could happen
uh uh via a club and
if the club really does produce benefits
then
the next city over the next community
over
is going to have an interest in starting
it also because the benefits are
are substantial and at the same time a
club
can grow in the participation
participation rate in a local area could
grow
so clubs can grow in size and they can
replicate into new
locations and that sets the stage then
for viral growth
and that it would be how the the
societal transformation and the use of

new systems could spread to near global levels

over a period of the 50-year horizon

so uh clubs are designed to basically uh

out compete native systems by native

systems i mean just the

you know the the the

legal economic governance systems

that we are using today

these new systems will obviously be

designed for a purpose and that purpose

is to benefit and

improve the quality of life and the

security of

of participants and uh if

if if they achieve that

they will out compete native systems for

the public's attention

engagement trust and respect so that is

the that is what they should do and

uh and if they do then then

the their use would would quite

naturally spread

in a viral fashion now the clubs

i just want to say about the love the

club level so we're not talking about

say a family making changes that's

like this isn't really family focused

the whole project

it's not really focused at the business

level like a particular business

a new a new business model for example

it's not

focused there and it's not focused at

the city level so a new city charter or

a new

new way the city can operate to be

uh to improve somehow it's situated in

between the city level
and and the kind of family slash
business level
right and i think of that as the
goldilocks level uh
if a club was smaller this i would say
this this level this in between level
the middle ground sometimes i call it is
a
is the smallest level of organization
that would allow a club to be viable
testable self self-replicating and
effective
so anything smaller than a the club
level uh
you could imagine say if we're just
talking about a business well then that
business has to compete in an
environment
against other businesses to survive and
if those other businesses are
you know not on the same in the same you
know doing the same thing on the same
program they might have advantages
that the new business model might not
have
at the same time if we went to a higher
model say a city level uh trying to
change things at the city level well
that's suddenly
uh really expensive and difficult
especially for a large city
so there would be a lot of political
blowback for trying to change things
dramatically at the city level
and the club level is such that
that you could implement you could you
could implement new systems

without having to deal with uh
the need to to win elections
or uh uh you know get approval from any
public body right so this is
so a a thousand like-minded individuals
can just start a club
and no need to no need for anyone else
in the community to join in
they can do it all on their own just as
is so that's the idea
is to reduce the barriers to to to
participating
in a club in in new systems and uh
and have it be such that the that that
new systems could
virally spread over time if they if they
show their benefits
okay so just to make the point then when
i'm talking about societal
transformation the new designs that
we're coming
up with i'm talking about the designs
for a club not the designs for a city
not the designs for
a county not the designs for a state or
for a nation but i'm talking about
societal systems designs for a club
and the and the club will then
demonstrate to the world in field trials
whether this whether these new systems
work well
as as advertised or not
nice good point because sometimes when
we're talking about these complex
multi-scale systems it's good to make it
clear that we're talking about
policies affordances capacities that are
at a certain level of analysis we're not

invoking a nation state or something
like that

no we're we're doing something we're
working at what uh

a workable level you know this is
focused at a workable level this can
happen there's really no reason it
couldn't happen

and uh obviously getting the initial
funding for the project is a big deal
but once the initial funding happens i
don't see why this could not move
forward

exactly as or you know close to close to
what is planned here

um yeah next slide

uh is this the endgame vision 18 is that
where we're at

yep yeah okay so um

you know where is this all leading i've
already said there's a 50-year horizon
uh aimed at at societal transformation
on a near global scale

but but the real goal obviously the real
goal in this is

is for society's

humanity to to use

systems that are beneficial to them that
are most you know most beneficial or
widely beneficial to them i believe

that i should say there's there's good
reason to expect that a well-designed r
d

program that engages local communities
and the science

world could come up with designs that
are

substantially better than our current

ones i mean if you
if if the goal is to uh is to
increase quality of life and reduce
uncertainty
if that was how you were judging the
design of societal systems then i think
you would have to say that the current
systems are
are failing they're they're
terrible in the sense that we now face
uh you know enormous
stress and strain due to climate change
due to biodiversity loss
due to due to a host of interrelated
problems that all
all of which were preventable all of
which have been talked about in the
science world
and in other communities for the last
several decades
and all of which had we taken action we
would not
be under the risk that we are right now
so and
so if nato if native systems have given
us this extreme a world of extreme risk
then i think it would be fair to say
that native systems are failing humanity
and it would be hard to imagine that a
that a
sophisticated science-based effort with
some of the
best minds in the world you know focused
on this
this topic could not do far far better
than current systems so
just to connect it to one note from game
theory it's that the strategy has to be

resilient

along the way so it needs to have that
kind of like

introductory dynamics and getting off
the ground and bootstrapping or
not relying on lateral connections
amongst clubs it has to be sort of an
end in and of itself

and then it also has to be a resilient
uh

mixed strategy at 50 percent it has to
be a resilient

dominant strategy and so flexibility to
play different strategies at different
um phases of the uptake is critical
that's why you've laid out a timeline as
well

yeah yeah um and that's also

you know kind of related to that to your
point is that

there could literally be hundreds or
thousands of experiments going on around
the world in different communities of
of different system designs and
different asp different aspects of a
particular design

so the so you would want to be very
flexible

very adaptable we're learning as we go
as a matter of fact

uh and would up you know would change
the change of the strategy slightly as
we go and

change the focus slightly as we go
adjusting the circumstances

and and starting small and growing
organically

helps you to do that as opposed to

starting big and
perhaps failing big so in the end game
vision the idea is that in the future
humanity is using systems that really
serve humanity
uh uh as we've talked about many times
now
it's just um the view here is societal
systems as a cognitive architecture as a
way of
facilitating societal cognition and
obviously we would all like
our societies to be good at thinking
good at learning
good at adapting good at making
decisions so that we can all
enjoy the benefits of such so
the endgame vision is perhaps i mean
there's here's a couple of
scenarios for the endgame vision the r d
program moves forward we demonstrate
really
useful designs where that give
substantial benefits
uh to communities and the
existing native systems uh slowly uptake
those ideas and blend those ideas
into their into their systems and then
in the end
you know where you the native systems
become much better
so that would be great that would be
fantastic and another scenario
is that native systems fail to adjust
and change and up
uptake these new ideas they uh and if
that were the case
since the the new systems are designed

to
earn the public's trust and respect and
attention
um they would it would seem reasonable
that they would out compete native
systems for the
for the public's trust and respect and
if that's the case then
you know people would be using these new
systems and relying on them
more so than native systems so those are
just two scenarios but you could
you could imagine any num you know a
number of other scenarios that might
occur
from such an r d program but uh the
either way at least with these two
scenarios either way the end goal is
reached of
of of you know a large majority of the
human population
using systems that actually serve them
and serve them well
slide 19 yep
okay so a little bit more in clubs
so by the by the time the first field
trial occurs in the first
system and this could you know
potentially uh the field trial
the first field trial could occur maybe
anywhere in the world practically
but by the time the community
participates in a first field trial it
should be really clear
what the expected outcome is right a
community does drive its systems a
community makes decisions day to day
the of of what you know what kind of uh

you know about services about waste
control about
energy generation about all kinds of
things a community makes its own
decisions
but the general trajectory
towards uh improved quality of life
improved well-being
improved security improved economic
security
uh reduced income inequality
like all of those things
there should be very few surprises as to
what occurs in a field trial it should
be
there would have already been numerous
simulations that would have occurred
before
uh before our first trial first field
trial already uh
before it even begins there would have
been all kinds of
user testing that would have occurred
there would be all kinds of game playing
with new systems that would have
occurred
so by the time the first field trial
starts this should be a
you know this should be a trajectory of
exactly what
is uh what is predicted and so there
should be few surprises
for the field trials now obviously that
doesn't mean that something
you know there might not be hiccups or
whatever but by and large
there should be very few surprises the
community should know really what

what they're getting into how the system operates

and what the expected benefits are in each year of participation

second item is that that

while the initial field trials might be really focused at the individual club level as the

research program progresses there would be an effort to

connect clubs via networks so that clubs themselves are participating in some kind of cognitive

operation um cl as a community of clubs that also needs to make decisions to learn to adapt

uh to make decisions together so the the idea is that some of the same kind of approaches and technologies that are implemented at the club level could also be implemented at the network level

so that the entire uh the entire you know range of clubs and the individuals within clubs could all be um participating in some kind of optimal cognitive system right making good decisions and look

and and opportunities for learning uh next slide

um so why why why

why the why the club model uh i've already mentioned a few things uh it's in the goldilocks level that allows for replication and and and viral spread

but there's a few other uh advantages that i would like to mention

um first we're talking about when you think about bold change revolut violent revolution is one kind of bold change so we obviously are not talking about revolution we're not talking about any kind of violence whatsoever we're talking about improving the well-being of communities at every step along the process so this is non-violent and that's one part another part is that success is measured and success and progress are measured throughout the throughout the process so not only might you be doing better but that would that would be validated the through a variety of metrics you know there would be this is an experiment you know this is the science world is engaged in this ongoing experiment to develop new systems so data would be gathered and you and everyone would know that you're doing better if you're doing better right that we would know if they're working or not maybe that's a better way to say it fitness the evaluation of system fitness and the quality of life for community are central to this whole project again because the clubs are operating in that middle ground there's that minimizes political

opposition opposition

there's as i said any group can join a

it can form a club so that you don't

need a city's approval for this you

don't need a nation's approval for this

is this is just a a civic club so a

civic club can start nearly anywhere in

most countries or most democratic

countries

and so you don't need a you don't need a

vote and the re this is really important

because

i am i am speaking to the global science

community

here and i'm inviting the the global

science community to participate but i

don't expect that everyone

all scientists will will be interested

in this nor do i expect that all

individuals in the general population

would be interested in in participating

in a field trial or

or anything like that as a matter of

fact i would expect that most

most people you know most people in any

given city

will have no interest in this at all but

that doesn't matter

because again all it takes is a thousand

people

in in your city or your area to start a

club

that club shows benefits it quickly

becomes clear to others that this is a

this is a very useful uh

this is a very useful program incomes

are rising

incomes are stabilizing quality of life

is improving
the club feels more
empowered to make decisions that affect
its that affect its future
the there's a stronger sense of
community
um there's the ability to to a better
ability to deal with problems as they
arise and find
long lasting solutions so on
on numerous levels quality of life and
environmental
quality should increase information
should increase
transparency transparency should
increase democratic participation should
increase
and those are all really you know
attractive
attractive characteristics if if indeed
a the club
model provides that so uh
if that's the benefits then then again
the there's the idea will spread from
one community to the next and grow
within
different communities and it's not
necessary that
that that a large percent of the
population
jumps on board on day one it's just it's
just not necessary
there must be i i would argue that even
today if we looked around
looked around there would be a community
of a thousand people who would be
interested in participating in a field
trial

and the first field trial of course is
still some years away
the program isn't even funded yet so by
the time we get to the point where we
really need a community to
run a field trial i i imagine there that
will be quite easy
as a matter of fact i would expect
numerous communities to offer themselves
as participants for the first field
trials
um this just a few more points here but
uh
the city whoever in in whatever city the
first field trials occur they're likely
to have beneficial spillover effects so
uh you know tax revenues might rise for
example
uh other other spillover effects from
the club
might help the city at large so
you know maybe even cities would welcome
such a thing if it's going to improve
the quality of civic life
uh for those who partici for those who
participate and for those who don't
participate
um and this this club model is opposed
to
say going big to changing the for
example the u.s
economic system or the global economic
system on
you know in a on day one or in a short
amount of time
this uh this local global viral strategy
is far less cost
costly far less risky and far more

flexible we're talking about
starting small bootstrapping or growing
organically
and that all can be done on a
on a practical budget as we'll talk
about
shortly and a lot and as i mentioned it
allows for parallel testing
uh so numerous field trials in numerous
countries and or cities could be
happening all the same time
and as far as cost goes what you know
what would be the cost well it's a
little hard to say because the program
isn't really
developed yet so but this is just a kind
of a guess on my part
but i'm i would think that in the first
few years
the cost might be less than 10 million
dollars a year to get this started
and then in in sub subsequent years
still in the first decade
the cost might be on the order of tens
of millions of dollars
now that is you know in a certain in a
certain sense that's a lot of money but
in
in a certain world that's a lot of money
but in another world that's peanuts
so this is not this is about the same
cost as a
modest size or a good size startup uh
you know
so that this level of funding is
available
in the world uh it's this is not a crazy
cost

and uh i i argue that
you know with this modest cost we could
really change the direct trajectory of
humanity here
okay that's it for part two
oh that's it for part two okay all right
so uh so
we have we will continue this uh talk
about part two in our next
jam board so i'll add the new link and
we're now on
slide two of jamboard three okay all
right good
um uh so the slide two
is uh is just where you can go for more
information so
if you'd like to know more about the
proposed program if you would like links
to all the articles
and links to all the previous videos
that we've made
and that i've been involved with you can
find them all on
[Music]
principledsocietiesproject.org
all right slide three um so there's a
few remaining topics from uh part two
that i'll talk about now
um some opportunities outside of clubs
uh
a little bit about fitness metrics and
then some final thoughts on part two
yep so we've already talked about
opportunities in clubs which could
include for example
increased income um and uh
higher quality of life and a bunch of
other

material benefits material and social
benefits

but there's also some opportunities that
could occur for those outside clubs
and as i mentioned clubs sit at the
midway level

between the kind of family slash
business level

and the city level and both of those
levels are already well developed and
you know there's an enormous amount of
revenue attached to both of those levels
the middle ground is not yet organized
but as it organizes i would imagine that
revenue opportunities are you know
similar to those at the city level and
those at the

at the family slash business level

so uh the i the the the

the r d program would help to do to
organize this middle ground the club
level

and some of those there's a little bit
of evidence that that such
organization is already happening for
for example uh is it called

what is the app called the neighborhood
app

it's the name escapes me right now but
there's a

there's an app that is widely used and
is used by millions of people in the us
at least

about communicating within your
neighborhood and sharing information
within your neighborhood

right so that's already a big business
that that app alone is a big business

cities are also developing community energy supplies community solar and that's just another way that this middle ground is organizing and if the r d programs goes forward there would be a tremendous amount of organization happening within this middle ground so that opens up opportunities for consultants for example most of the work and developing the r d pro program the the tools and technologies needed for field trials for new systems that would mostly be open source you know publicly available uh knowledge and and information but it could be that that businesses consulting businesses or other businesses could provide services to new clubs could provide these could be training services they could be you know like internet services or any other kinds of services that would help clubs get started and uh i imagine that could be a big business as we're as we're as this moves forward maybe a good model for that is red hat incorporated which uh which uh provides services related to the linux operating system provide services to businesses related to linux linux itself is an open source product or it's open source technology and red hat which is a enormous company

i don't know if it's a i see a billion
dollars in tax revenues recently so
this is a very very large company
operating on top of an open source stack
in a sense and the same could be uh the
same operation similar opportunities
could occur for
for for businesses who want to work in
this consulting or service
providing space another opportunity is
for cities and other governments
because of tax revenues later
in i don't know if we'll get to it in
the next session but
maybe in the in the next next session
section of these discussions we'll get
to the uh the simulation model that i
published on
what i call the the lita framework local
economic direct democracy
association framework uh that is that is
that simulation is just illustrative of
a potential uh kind of a prototype new
um
societal design uh focused on the
economy focused on an economic system
but in the simulation that occurred the
simulation was for eugene oregon
using data census data for eugene oregon
and uh
the simulation occurred over a 28-year
period but by the end of that 28-year
period
um let's see i
headed down there just a few words about
what that provided but
um
billion in tax rate right right which

was more than double the initial tax revenue so potentially this is a you know could be in a sense kind of a windfall for uh local governments city governments and community governments uh for for tax revenues and then finally there's an opportunity for large investors and that uh you know we are looking at a future world that is going to be very stressful because of the the implications of climate change the effects of climate change uh biodiversity loss and social other social environmental problems so if it happens that uh that there's a shortage of good investments to protect your money the club model and the club process and the r d program might become they might be seen as safe havens for parking money until things uh you know improve and if that's the case i could imagine that some institutional investors who who desire to make social investments or who see the wisdom of making social investments could invest in this whole process and of course that would provide funds for developing the developing the club model and helping the club model to spread and um

and grow um
yeah who knows who knows what is going
to happen in the stock market as
clim the ravages of climate change
approaches
and maybe the maybe this maybe this club
model idea could really actually help
large investors by
providing a safe haven uh for their
investments
and i'm talking social investments i'm
not talking um
highly profitable investments that some
might be used to but i'm talking about
uh social investments where it's
invested in a low expectation of
a profit
okay slide five
that's cool uh okay so just a little bit
on
fitness metrics we've been mentioning
them all through
these talks i just wanted to say that
they serve several purposes
for example they are they are this
fitness metrics the information gathered
about fitness metrics so for example
just to name a few fitness metrics you
know could be uh
family income could be education levels
it could be
water quality air quality i mean you
name it that would be
you know potentially uh hundreds or
thousands of
of of topics to collect information on
so this in this information could help a
club assess its own status

how am i do how are we doing you know we
need information to answer that question
how are we doing well we can say that
we are doing blank you know we're doing
so well or
so poorly or whatever the case may be on
education on environmental
quality on economic security on all
these
different issues um second is
this information helps clubs to solve
problems
we have for example the
how do we how do we maybe the problem is
focused on economic security how do how
do we solve this economic security
problem well we need information we need
to know
uh how we're doing right now how are we
doing last year
so that we can know we can make good
decisions for solving these problems
that we're facing right now
metrics are also useful for assessing
club and system design
again clubs are built for purpose and
are supposed to function
successfully for purpose and achieve the
aims that they were designed to achieve
and all of that to assess club fitness
and to
to assess how a club how a club design
is doing requires information
another use of information is just
informing members so
everyone has questions about you know
just day-to-day questions in their life
you know what career should i choose or

or you know
what where is it where can i buy fresh
strawberries today or whatever
you know whatever information that
individuals need in living their
day-to-day lives and running their
businesses day-to-day
the information gathered by a club could
help them to make better
decisions just in their own personal
life
and then lastly the data assessed would
just be a windfall for basics
basic and applied science this were you
know in a sense we're talking about a
repository a a large repository of
longitudinal information that follows
individuals and families and businesses
through their life and uh and how their
how they're doing their quality of life
as as decisions are made and as the
as the as the club makes decisions as
they make individual decisions
um that information would just be gold
for
understanding uh you know societies
better understanding clubs better
understanding health better
you know than that information there's
only a few small
repositories like that today so we're
talking about just
a really a gold mine for for science and
a basic science and applied science if
this moves forward
obviously there would be you know
personal you'd have to protect personal
data in all of this but

that could be done and just a few words
about how this
these fitness metrics and data this data
gathering system
would be different from what's already
done well you know
already for example the census collects
information
and the sustainable development goals
sdgs
uh are proposing a you know a collection
of data at
the national level um that would be and
all of the all of that data for the
census and for sdgs
it touches on some of the same topics
like quality of life
and education and economics and so some
of the
large topics are similar but the
but there would be many many differences
between this kind of data collection
and uh and the existing programs
so just a few differences this would be
a larger and more detailed set of
information
it'd be collected and managed at the
community level not at the national
level
it would be about local conditions so
specific for local conditions rather
than say national conditions
forecasting other kinds of models are
integrated into this repository so this
would be a you know
an intelligent uh the aim is here an
intelligent data gathering system
it would be part of uh you know

forecasting
metrics into the future and things like
that uh
and the obviously the club is designed
to
to serve a different set of purposes and
in particular purposes related to
community cognition societal cognition
lastly clubs would collect information
on a far more frequent basis so
uh you know some some survey questions
are asked every
few years sometimes sometimes every year
sometimes every few years sometimes
every five years
and the sdgs i think are collected
mostly on a
annual basis so uh this is on a perhaps
you know some some data would be
gathered minute to minute or hour to
hour or
day to day so there would be a far
richer
and higher quality data set that would
be
that communities would generate yeah and
the basic idea is the community needs to
know what's going to happen to me like a
community needs to know what's going on
where did we come from what's going on
where are we headed what should we do
and to answer those kinds of questions
requires rich data
and that's exactly what would be
proposed here community could
collect rich data about itself so it
would know how it's doing
and what's going to happen if i could

also add one comment there

with active inference type models we can have a rich uh or deep generative model even if we have sparse data collection initially and then we can ask what kinds of collection of what type of data or what type of modeling or what perturbation would make sense given our generative model or how can we differentiate between two generative models or two hypotheses at play

so the richness can be specified up front we don't need to just grab what we can see in terms of data we can actually be principled in terms of what information is pursued

yeah yeah yeah as we've already said part of the

the concept behind active inference is decision making under uncertainty and obviously

we're talking about a community that doesn't know its future and isn't sure what's going to happen next because no community understands its future very well and and especially in today's world is very uncertain

about what's going to happen next will there be enough food for us next year will there be enough clean water for us next year

i mean these are some big questions but there's a million little questions and so somehow a community has to make decisions under uncertainty and that's

really
you know what the point of active
inference is is how that can occur
um and and then you uh touching a point
that you just raised
daniel is is uh
you know active inference is a is a is a
is a
is a way to make steps
in cognition right it's a flow of
information
processing decision making action
information flow decision making action
so
it naturally lends itself to an ongoing
process of learning
and also a multi-scale one so there's a
multi-scale learning here just like you
said some are
one-year questions one day one hour
there's multiple ticks to these models
that we've seen
nested models at different time scales
cool right right right
i also want to mention too is when you
know when we talk about models sometimes
we might be thinking
compute say computer models and that's
fine you know computer miles
computer models can extend the the
intellect it's extension of the human
capacity for processing information
but you can imagine too that individuals
can
individuals do process information they
they learn
themselves about their world and what
might happen next and what they expect

if certain decisions are made
so you know part of part of
the flow of information and the process
of processing of information at the club
level is
is to ask people what they think will
happen
if we do this what do you think will
happen if we
enact this rule or you know
put our put our uh funding priorities
over on this topic uh what you know what
it what is going to happen if we do that
what's going to happen tomorrow what's
going to happen
next year what's going to happen to my
grandchildren if we make this decision
today
i mean these are questions that humans
can can
take a stab at you know can answer in
their to the best of their abilities
and we might also imagine that certain
humans are quite good at it
uh you know just because certain that we
all have talents in different areas
so uh you know this is not just about
logic and computer models this is also
about engaging the
human spirit and the human heart in in
in directing uh
path towards a better future and making
decisions that lead to a better future
both both in the near term and in the
long term
nice
all right so some final thoughts to part
two

um uh i've i've already mentioned
that it seems already that there's
enough public support for this
because it doesn't require a lot of
public support to get off the ground
again just uh you know as far as the
science community goes i imagine this
could move forward if just
uh you know dozens or a hundred or
something like that scientists were
really interested in this topic i
imagine it could get started
and if just one community somewhere in
the world is interested in
perhaps doing a field trial of any new
system once it's developed i mean that's
that's really enough
so we're talking about a tiny fraction
of the world population that would have
to be involved in this to get it off the
ground
and just like any new technology uh
the iphone or you know like like it
doesn't
it once you build it and you demonstrate
that it works
and it brings it reduces people's
uncertainty about what's going to happen
next and it empowers them to make
decisions
that improve their quality of life and
and improve their security going forward
then um you know the rest is history i
think that's that's how i imagine this
would work out is you just demonstrate a
better way
and it has such benefits that it just it
just spreads

so to get it off the ground just a
fraction of the world
would need to be involved the fraction
of the population or a fraction
of the of the of the global science
community
but but once it gets rolling then i
think people will actually jump out of
the woodwork to be a part of this
because it's
like who doesn't want a better world
that is demonstratively
better than what we had what we have now
all right so one other uh so there seems
to be enough
support there's there's groups uh
various groups working on similar kinds
of uh
you know topics about say for example
transformation and other
topics like that so i it's hard for me
to imagine that
that there wouldn't be enough support to
move this forward if there were funding
funding obviously is the is the big
thing to get over right away but
if there were funding i think this would
go fine uh another
another question mark i have at the
bottom is
you know it's a good question would
transformation actually be easier than
reform
i mean in free form you're asking people
you're asking corporations for example
large corporations to change their
behavior even to change how they measure
success of their

endeavors for example by including more social metrics in addition to just revenue metrics you're asking power you're as we'll talk about soon enough you're also asking people those with those with concentrated power to distribute that power to the rest of the world and uh you know you're this is a big ask your reform requires some big actions by big players and that's great i mean reform is absolutely necessary and every effort should be made to move reform forward it's it's we won't we might not survive without reform it's especially good for the short term to get things uh you know improve conditions and improve our trajectory in the short term but in the long term transformation is necessary can i ask a question yeah um you mentioned funding being a rate limiting step so what would you spend money on oh as well as a couple slides ago i mentioned that it might cost about uh whatever maybe less than 10 million or around 10 million a year for the first few years but there has to be you know a variety of things would have to happen just to get the program going you know there would have to be conferences that would

have to be
some kind of staff to handle the you
know the
the program that would have to be
you know some early studies done like
the first
studies would have to be funded a lot of
the funding would actually just go
towards studies
but uh studies about what the public's
opinion of transformation is what the
you know
what how transform how the idea of
transformation is seen for example
um simulation studies of
potential to showing potential benefits
like there would just there would be a
slew of work that would have to occur
and and
we're talking about work by
professionals of scientists and
otherwise
uh and uh you know that that just costs
money
the 10 million is actually not much so
it's a small that's a small
team to get things off the ground and it
would cost about that
there could also be funding for arts
projects funding for
literature funding for stories about
what a transformed world would look like
still you know works to to to
bring the idea of the potential like the
potential of trent
to tell the public that we actually can
there's a path we can
transform we could we could live under

different systems

and what would that be like what would

that look like what would that feel like

what would we

you know how would that be you know

that's like a it would be

lots of art things and music things and

literature things and

media things that could support that in

addition to all the

science-based uh efforts so i don't

think

you know there's there would be a few

things you could do without funding

there could be maybe a few studies that

you could kind of squeak by

and the spare funds or something but

without without

substantial funding this i don't think

this would really go too far too fast

all right all right well that brings us

to

our our real topic part three

nice uh

so part three is focused on on design

and when i say design

i mean high level design concepts not

not nitty gritty how would you do this

how would you do that but

high level design there's a few things

that i talk about in this paper the the

paper itself is a narrative on the

topic of de novo design you know

designed from scratch

of new systems um obviously as we've

said

many times now the society is views it

as a super organism and its societal

systems as a cognitive architecture
so in that sense the goal of design is
to facilitate
cognitive societal cognition through a
through a better architecture i
i talk i i spend time on a few topics in
this paper
an attention communication self-identity
power
and influence are all discussed and uh i
also talk about uh prototypical
uh societal system design the this is
the leader
framework that i mentioned just a minute
ago i don't think we're going to
get to the leader frame i'm sure we
won't get to the leader framework today
and there's a lot to say about that
because that is
as far as i understand is still the
first simulation
in the science literature about how an
integrated system
might function so there's much to say
about that
uh and and i don't mean to say that it
uh you know the whole
the whole r d program is aimed at the
lead
infrastructure it's not a leader
framework it's not but
the the those concepts could be useful
in
developing other ideas for societal
systems
and at the very end of the paper i
highlight some design considerations
so we'll get to those maybe in two at

least two sessions two or more sessions

here

cool slide eight yep okay so um

i just wanted to say a few more words

about you know

reform versus transformation so uh there

as i mentioned there are loud voices

at the u.n and even in governments

national government state governments

local governments

and certainly in the general public and

certainly in the science community

calling for bold really radical change

to avoid this

cliff that were hopefully

not going to fall over here related to

climate change related to biodiversity

loss

and a host of other uh social and

environmental problems

and um you know just to just to kind of

emphasize how how dangerous

our situation is it is not clear if

humans will go extinct with all the

other extinctions that are happening

and i don't mean extinct in a thousand

years i mean like will there be

will there at least will there be a

civilization any any sensible

kind of any any civilization that we're

used to

will that exist in who knows 10 years 20

years 50 years 100 years you know like

the

our our future is highly uncertain

because we have not addressed

deep problems that have been with us for

for many decades we've allowed them to

grow worse and now we are all at risk at
risk of just
lower quality of life or risk of mass
die off of humans and at risk
ultimately of human extinction
so there's lots of voices calling for
bold change
and that bold change can be seen in a
couple of ways more
the typical way to see it is reform of
uh isolated societal substance
systems like uh there's a lot of focus
if you just read the news there's a lot
of focus on
green energy for example using solar
powered energy and
things like that uh better trans you
know electric cars rather than gas
gasoline engines that's all great again
all of this stuff is wonderful but i
would
i would offer that that there's
something
really missing from this kind of
assessment and that is the larger
picture
the the the the the meta level
uh they're sometimes called in the mlp
of the multi-level perspective
gills it's called the the landscape
level or the regime level
and i would say that we have to address
that level if we're going to really
solve our problems if we're really going
to develop
if our systems are really going to act
well as cognitive systems then we have
to address

the systems level the parameters
and characteristics so right now the
focus
of all all the work that's being done in
sustainability and this bold
the idea of bold transformation almost
all of it is focused on this kind of
meso in between level uh you know
the say the energy the energy
infrastructure or the
transportation infrastructure and this
paper
in this whole series instead is is is
really focused at the
at the landscape level the the large
level
the whole transformation the whole
systems level okay
i just wanted to make that point clear
and okay so now now i
you know we're we're already in part
three of the series and i've
i've i've listed six systems that i
that are six you know meta systems or
landscape level systems that i've that
i'm interested in economic political
legal health science and education but
in
in the first few pages of this uh paper
i
i finally say what that actually means
so uh so uh the
i'm basing my definition of a societal
system on what's called a function
system
and the more there's citations there by
roth and schultz and others about what a
function system is

comes out of social systems theory but
they list uh

three characteristics of a function

system it has to have be observable

and not a subsystem of other societal

systems so the

economic system is a you know entity in

and of itself not a subsystem other

other systems

um involves a communicative system

whose operations refer to a society and

manifests the functional perspective at

work

at work so uh roth and schultz

identified ten

function systems and uh

i would say that uh six of these i'm

calling the

the overarching systems the economic

political legal and so on

in in parentheses there i have their

functions so the

function the one word function of the

economic system is distribution

the one word function of the legal

system is standardization the health

system is restoration

and so on the education systems

formation

there's a four other uh of function

systems that roth and schultz identify

uh art religion sport and mass media and

i

i do talk about these but i talk about

them more indirectly

so that's what i mean by a function

system

and and i also use the word

or that's what i mean by a societal
system and i also
read the word institutions in throughout
the series

and it's worth spending just a little
bit of spending a little time
describing what i mean here um i use
that in

the term in two senses one is just
concrete organizations for example a
attacks or attacks uh institution
inside of a governance system for
example but the second meaning is is
more broad it's uh

[Music]

more more along the lines of a social
institution

and this is kind of interesting because
um

grabner and gobarney
define social institutions they give a
little like a little

blip there codophile systems of social
structures and particular norms and
rules

that incline individuals to act in
specific ways

and uh interesting the idea this is you
know they've developed these ideas apart
from active inference but the

but the language is not so dissimilar
because they talk about institutions as
constraining and enabling people to act
and

coordinate in complex and uncertain
environments

they stabilize the expectation of of
individuals about the behavior

of others and uh thus the institutions allow individuals to offload some of the cognitive burden that that they would otherwise be required to compute if there's some kind of social norm and i expect you to behave in a certain way under a certain certain stance

i don't have to spend a lot of thought figuring out what you're going to do next

as i go about my life you know i can i can just assume that you're going to act according to the norm and then that's a whole lot of work that i don't have to do

to to forecast what i think you're going to do

so uh institutions allow individuals to offload some of the cognitive burden and uh and and anticipate the state of play

and the beliefs of others so obviously that's that's related to active inference right

active inference is uh uh at the you know when we talk about uh uh um of forming you know structuring a society or structuring a society to reduce uncertainty

one way to do that is to develop a set of norms that this that everyone agrees to where everyone follows or mostly follows so that the uncertainty about what's going to happen next is reduced so so

active infra developing institutions if
they're quality institutions actually
helps them to reduce free energy and
expected uncertainty
i i think there's a few dimensions to
this and it will be fun to unpack
with what does it look like to have the
active inference institution
what does that mean as far as the
operating procedures of the
internals of the business versus how the
stakeholders interpret it how it's
communicated
so how it actually plays out will be
something we're just exploring
yeah yeah and when i say institutions
i'm you know i'm talking uh
widely so financial institutions uh
you know social institutions cultural
institutions
and so on um and i'm i'm suggesting that
they can
the institutions can be viewed through
the lens of active inference and may be
improved
by understanding you know how human
cognition works and how group cognition
works
we might be able to improve our
institutions
um just some reminders from from our
previous talks
about the goals and purpose of the r d
program and of
and of the purpose of an organism itself
and thus the purpose of a
of a societal system that serve that
serves

a group of organisms the society
so the the goal of transformation from
this view
is to develop and implement systems that
can best facilitate societal cognition
so that's the goal we have metrics to
assess whether
a new design is better or worse than
some other design
and importantly
if if we weren't thinking about
cognition if we were just thinking about
conditions
we might say the goal of transformation
is to
you know improve education levels or
improve diets or improve
quality of life in some way or improve
economic security or any
any number of things like that all of
which are fantastic are good
but once we bring once we bring this new
idea
of of of societal systems as cognitive
architecture and
into play into into play then we're
talking about
um the the
the conditions such as clean air or
clean water
being the outcome of a decision-making
process
so the focus starts to shift to the
primary
uh you know the the primal goal of
quality cognition versus the secondary
goal
of what quality cognition brings quality

cognition at the
societal level brings clean air and
clean water and low
uncertainty and you know higher
you know good education and freedom and
you know any number of other
characteristics conditions you might
want to list
so it is important that you know this is
this
the by looking at uh societal
uh systems as a cognitive architecture
suddenly we have a much
richer understanding of of not only how
a society works and how decision making
in a society works but also
we have new metrics by which we can
measure the success of a societal system
right
and so this would include cogniti
cognitive
metrics data metrics information metrics
processing metrics you know all kinds of
metrics
so that's a little bit on on the goals
of transformation and the goals of the
program
and a little bit on the purpose uh the
purpose of the program and the purpose
of a
of a new design right i just want to say
that
the purpose of a design is both
aspirational and functional
we've already talked about design as the
last step
in the process of world view leads to
purpose leads to fitness

fitness metrics and then finally to some
kind of system design
so in that process in that process
obviously uh purpose is uh
is functional like you know everything
else practically everything else depends
on purpose
you know or many things depend on
purpose but beyond that
having an explicit purpose for a club
say it you know like for a particular
community
a clear statement of useful purpose
can actually help in and of itself to a
shared purpose can help to unite
individuals and focus attention
and uncommon causes and encourage
cooperation if we all know
why we're doing this together if we all
know why what we're doing and why we're
doing it
and that and the and that purpose is to
improve
our collective quality of life and
improve the environment
well then it's going to be a lot easier
to get people to
come together and talk and and get
and you know unite behind a shared
purpose
uh so in this sense a purpose itself
could aid societal cognition right
societal cognition depends on
cooperation depends on trust
so just having a just having a purpose
can aid in societal cognition
and and you know like as a as a
counterpoint to that

what is that what is the purpose of our
current society you know we have these
native systems of governance and
economic systems you know so what is the
purpose

what is their purpose right what is the
purpose of an economic system
you know i mean we could we could debate
that right but

but there it's i would say that it's uh
that if you just take things at face
value

one purpose would seem to be to promote
revenue and sales and consumerism and if
that's the purpose

if that's part of the purpose of an
economic system then why would
you know it's hard to get behind that
you know it's hard for a
uh for for a whole society to say oh
that's a that's wonderful let's get
behind consumerism

because it's so destructive

okay um so a little bit on goals and
purpose and on to the next slide

uh this is just a quick review of active
interest maybe i don't even need to say
this because we've gone over it a few
times but

essentially uh you know the short view
is that we can

we can view action action cognition
as an uncertainty reducing process
so that's that's what what this is all
about in a nutshell

and uh the the uh through
via via the active inference mechanism
we

take in observations we update our
models we learn
we take actions that can reduce our
uncertainty either
either either explorative actions that
bring in new information epistemic
epistemic actions or goal-directed
actions that actually
immediately or in the short term or act
to improve our situation
so it's all about reducing uncertainty
and especially
it's about reducing the uncertainty of
of
essential variables and this is of
course uh you know this is a rich topic
in and of itself what are the essential
variables of a society
i mean they you would imagine that in
some way they would include clean water
and food and you know sufficient food
and
clean air and shelter and all those
kinds of things but as we talked about
earlier
in the last discussion there's we talked
about
maybe eight or so core human needs that
include affection
and freedom and a bunch of other core
human needs
so we're complex organisms and we have
complex needs and these
these essential variables the variables
that really matter are themselves
a complex set of variables and if we
and if we focus our attention on
on addressing this question of

uncertainty about essential variables
that almost means by definition
we're doing what i what i like to call
solving problems that matter
problems that matter are the problems
that really relate to the essential
variables about how
are we really doing how are we going to
do tomorrow how will we do in 10 years
yeah so so
i just we'll just read the last the last
sentence of the last
block of text i have there this is by
bad
and they just they phrase they're
speaking about active inference
and uh the act of influence as it plays
out at the
social institution level and they say
that the elegant idea that we can
operate together to minimize collective
uncertainty
stands to cast a new light on classical
social phenomena and that's about it
in a nutshell that's it and to draw one
point on that
in contrast to for example reinforcement
learning based approaches or reward
maximization based approaches
it's almost like you have the gaussian
distribution it has a mean and a
variance
and the reward learning is like let's
shift the mean
up as much as possible i mean isn't the
average income
right now in us dollars higher than ever
well maybe but then if you think about

what the dollar means and how the dollar
is distributed
then the average doesn't really tell you
too much about the distribution
so active inference is about reducing
your uncertainty
on a specified generative model so
somebody might think i think the
distribution
we might want to see this type of
distribution this shape or this kind and
that's a discussion to have
and then we can reduce the variance
around that
and let the mean fall where it may so
it's very different to have a reward
highest as possible because having high
expectations
but in unstated and often unformalized
variants can lead to some ridiculous
outcomes
like the saint petersburg paradox and
other uh
investing fallacies let's say yeah good
point
um uh so so your point was that uh
uh reinforcement learning is aimed at
not
it's not really a necessarily a
probabilistic uh
approach and and if you and you will
have to
really ought to include the distribution
of whatever
targets you're looking at in order to
get a better idea of you know
of what's a good decision and what's not
a good decision

absolutely another another difference
between
rl and active inference is that the
programmer that whoever is designing the
experiment or the whoever is coding the
robot or whatever
has to decide in advance what the what
the
what the rewards are for certain actions
so
you know you fall through trapdoor
negative 10 points you you
you you get the you win the chess game
you know positive 10 points or whatever
the game is
and uh active inference allows the
the agent to learn goals as it goes
uh learn the world as it goes and make
to change its
change its priorities as it goes which
is
which is exactly what humans have to do
we have to be learning
as we go and changing our perspective
and changing our what our goals are
and how what we're valuing we have to
change that on the fly
in order to maneuver in a highly
uncertain world if i could add one more
point on that it and this was expanded
upon by
uh noor sajid and philip ball in our one
of our previous discussions
that relationship and the learning of
the cues in the world can be basically
semantic
and arbitrary so sugar isn't just always
simply a reward for the human in the

environment

like right now i don't feel like having

sugar so it's not part of my preference

vector

so with active inference type models we

can separate out okay

what's on the table in terms of our

current affordances

how about future affordances what are

our current preferences and how sure are

we about them

and all these parts of the actual action

plan

that will help us align on shared action

rather than thinking that if we simply

line on

shared thought then all of a sudden

we'll know what to do because that's not

even true

right right good good

okay so um so

um we're we've been talking about

societal systems as a cognitive

architecture all through this

but what is maybe it's time we actually

gave a little thought to what cognition

actually is right um

so uh lyon lists 13 cognitive tasks that

he

that that he says qualifies qualifies as

universal tasks or

capacities and uh it is ra you know it

is

this is this is part of the world view

that we we talked about in the first uh

in the first discussion in that the

all of life is intelligent all of life

is engaged in cognition

all living things are engaged in
cognition and these are some of the
you know some of the tasks and
capacities that what we mean by that
so this these hold true for a human
certainly

but they also hold true for bacteria or
slime mold or plant or any
any other living organism

i i don't know if i need to go through
all 13

one by one but i'll just kind of jump
around a little bit so

just number one there has to be some
sort of repulsion or neutrality or some
kind of judgment of what's good and
what's bad what's useful and what's not
useful

certainly to cognate you have to sense
you have to bring in information
there has to be some kind of memory
storage or some kind of recall

there has to be some kind of learning
some kind of problem-solving behavior
there has to be communication with other
agents of the same kind or different
kinds

there has to be anticipation certainly
we if we're going to
make a decision about our future we have
to anticipate what our future might be
if we do or don't do some kind of act um
there's attention there's a million
things to spend our attention on as a
human

and what we spend our attention on will
govern

in large part the quality of the

decisions that we make
and the last one i'll talk about is
self-identity
so you know if i say i'm going to the
store
in that sentence i'm talking about this
particular body right
and my self-identity is focused on this
particular body
but i can also talk about myself as a
citizen of a community as a citizen of a
nation
of a of a as a part of an ecosystem
as a part of the biosphere so we have
the capacity to self-identify
at numerous levels i call that extended
extended self-identity you know not just
ourselves maybe our family maybe our
community maybe our environment maybe
our local environment maybe our global
environment
and maybe of human civilization we're a
part of human civilization
and as we'll see shortly how we
self-identity
how we self-identify governs to some
degree
the kinds of decisions and the kind of
kinds of behaviors that we display
so uh so these are these are we can
think of these as the
you know occurring in every organism and
the next slide brings the two ideas
together the idea of function systems
and the idea of these universal
cognitive tasks how do they relate to
each other
and i'm offering that that in a sense

these 13 cognitive tasks are really you could think of them as building blocks for the higher level functions so for example uh an economic system say in the in the function view an economic system might involve all 13 tasks in one capacitor or another a education system might involve all 13 tasks but might particularly involve the tasks about learning for example so so we can think of a lines list as containing low level cognitive building blocks and function system as being the higher level functions built on top of these or from these and then we can think of active inference as the multi-scale process that connects the two and these function systems then provide a cognitive architect architecture that provides a society with a with a cognitive flexibility that is a society needs to make decisions it has these uh you know six or ten different systems different uh function systems that it can make decisions through that it can learn through and there would be certain kinds of problems that are maybe better suited for say an economic system or a governance system or something and other kinds of problems that might

be better suited for
a legal system or education system but
maybe some problems
really have to engage multiple function
systems
so the fact that a society has these
various uh systems of cognition
it gives it a certain cognitive
flexibility flexibility
and provides a certain kind of
redundancy and
and then obviously raises the question
of what kind of balance are these you
know eight or ten function systems
you know how does the dynamics of that
balance play out
uh next slide okay in this slide i'm
just listing
the next four things we're going to talk
about and that'll be the if we get to it
that will be the end of this
our discussions today um those are
those are neuronal versus human
societies what are the similarities
what can we learn from neuronal
societies that might help us when we're
thinking about design
all of these all of these four points
are related in some way to the design of
new societal systems
influence how do we influence each other
and particularly
what is the role of coercion and
influence
self-identity and selfishness we we just
were talking about that when i was
talking about expanded self-identity
and we'll pick up that theme when in a

short discussion about
cancer what is cancer and then lastly is
power distributions we have
we have eight or ten or six or ten
functional systems
uh if one is you know if one of those
is unbalanced with the other two has too
much emphasis in the society than others
how does that affect a society
and then also within a system if for
example in an economic system if
uh financial power is is
uh you know concentrated how does that
affect a society
so we'll talk about that all that and
and then with the idea that all of these
pertain to the design of new systems in
some way
we can learn something from these topics
about how we might want to design new
systems
so slide 16 um so it's kind of
interesting you know
our what the associations and community
that happens inside of our brain to
for human to solve problems is not so
different in some ways about how we
engage with each other in in potentially
solving problems
as a society so uh
i'll just kind of read a little bit from
that first paragraph but
similar to neurons individuals connect
with each other in complex ways just as
neurons do
we exchange a variety of signals over a
variety of distances just like
neurons do at a variety of strengths

both neurons and
persons can potentially connect to a
very large number of others
each can share each individual each
neuron each person can share multiple
connections with others
and these connections can form meta-
level structures like
institutions or tissues or whatever the
case may be
neurons and persons self-organize into
modules
of tissues families clubs businesses etc
and neuron neuron and person-person
connections are dynamic
and context and need dependent the the
the
the neuronal architecture is quite
flexible
just as the human human social
interactions are flexible
associations come and go even if there
remains some kind of large scale
you know consistent structure and and
each individual is cognitive just as
each neuron is cognitive that's actually
kind of a new
new new result in the science literature
the capacity of a neuron to process
information
in and of itself so uh
you know obviously we want our neurons
to work together we want them to
associate with each other in a way that
helps us to solve you know our
particular problems
we want their signals to be strong and
good

so that information is moves throughout
the you know the
neuronal process as it should
uh we want to do we want our neural
neural architecture to

[Music]

re-reorganize itself as we need to as we
need to learn new tasks for example or
new topics

and all of these are this is also what
we want for societies we

if we're going to be part of this
cognitive architecture if we're

if we're engaged in societal cognition
we need to associate freely with each
other uh

join join the collaborations and groups
as we see fit

that's part of societal cognition um
our you know societal cognition in
particular the designs

of societal systems might impact any in
this bulleted list that i have but
one the first is the strength and the
quality of associations between
individuals

the richness and the extent of our
communication uh for example a
a design a societal system design might
have

as part of it a means of communication
so that

say a large group of people can get
together and collaborate and coming up
with a decision on some topic right so
the the societal system itself could
have

communication and coordination aspects

to it
temporal and spatial locations and
movements of individuals affect our
societal cognition
and and designs can affect that uh
quality of information
that is gathered uh and in particular
i'd like to point out
the motivation to engage in say economic
behavior
hopefully will be different than it is
in native systems where the
motivation is largely to you know
increase
personal uh income and wealth
um uh we would want
go ahead oh just one one uh i
was gonna ask what the differences were
in contrast but finish whatever you're
going to
go ahead and say oh well just just on
that topic
you know a new design would want to
employ
a set of motivations that is you know
very useful for societal cognition and
societal health
and certainly encouraging people to act
in a kind of a selfish
self-centered strict strict selfish way
for their own gain would make little
sense
you know that's that's like a design
flaw in our current system you know that
we're not encouraged to
cooperate and encourage to solve common
problems together
and the design of a system can deeply

affect what our motivations for actions
are
as a matter of fact a design ought to
make cooperation
so easy that it becomes the de facto
default behavior
it should that should be the kind of you
would think that would be the
how it how a good design would work is
it would just reduce the barriers to
cooperation and communication and
coordination because
th those are all key to um
societal decision making and learning
yeah uh other just a few others here
uh the design might affect uh
for example how well as a society's
beliefs aligns with its reality you know
for example
if if in a new design
there's a strong role for science and
there's funding
in you know funding in clubs for
scientific exploration
well then maybe it's more likely that
that new systems will be more aligned
with
the beliefs of people working in new
systems or engaging in new systems might
be more aligned with realities because
you know the best science has it best
as at its best is aiming towards a
better understanding
of of of of our world
um design might affect how independent
and
co-op cooperative individuals are as i
mentioned a minute ago

um also might affect how uh well
how deeply the level of critical
critical thinking is right how how well
does a society does a community
encourage critical thinking through
education and through
norms and through other processes but
obviously critical thinking is important
in problem solving you
can't solve problems if you're not doing
critical thinking you can't solve
problems well
and and also the capacity to plan the
extensive and the extent of the
anticipatory horizon
you know what a design might affect uh
how how much attention is paid to
various predictions made by
organizations and groups
um maybe it would be important to pay
attention to the predictions that people
made so we can understand
who is good at who is who has a skill uh
you know who's good how can we improve
our capacity
to anticipate what's going to happen
next
so in numerous ways in the same sense
that
that design design of a brain system
might affect
cognition within the neural communities
some of those same uh ideas can play out
in the design of
societal systems because you know
associations and etc etc
would be important at the societal level
i'll just point out as an evolutionary

biologist that selection has shaped

[Music]

on stumbles certainly and so several

patients of today

have benefited from developmental

basically patterning that has been

selected on for a long long time

many many many generations so real

evolutionary pressures

the kind that create resilient neural

systems at the level of societies

would require essentially

population-like dynamics

competing and the birth and the death

and the diversification of human

societies

so systems mapping is really helpful and

the toolkit transfers there for sure

but there's a key evolutionary bit that

neuronal ensembles have that ant

colonies have

that simply saying distributed systems

are like those two things

uh is is not there but in the long range

that's where it would be yeah yeah um i

i think i've said this before but

um but you know if we're talking about

how can society do better at cognition

be you know what kind of systems might

facilitate cognition

well then all we have to do is look at

evolution we look at

how the what what is the world doing

around us how does a brain work

these are you know the the the

information and the ideas and the

and the and the patterns that we see

throughout nature

of throughout nature can be look we can
learn from them
as to how good cognitive systems work
because nature is built
that nature is a good cognitive system
right there's so much to learn from
nature
cool um all right so
this is the second of the four topics uh
influence
how do we influence each other so we
have a group obviously the people will
be
there's problems to solve there's
activities to to
undertake there's uh challenges that
arise
and group members would try to influence
with one another in some ways
through communication or you know some
kind of communication
and turner the fellow i'm focusing on
here uh
his ideas come from uh he did work he
worked in uh
social identity theory self-catering
categorization theory and some other
related fields and he uh
identifies two forms of influence
now this is not this this is not coming
from you know the active inference world
but i'm
going to make the point at the end that
the two are
closely aligned with each other um
one time type of influence is persuasion
and that's where you simply argue that
your you know your understanding of

realities in some way better than other understandings or your path of action is in some way been more beneficial than other paths of action but already then we're you know we're talking about aspects of of cognition in the human brain or active inference because we're talking about some kind of uh testing of a model you know of of looking at potential paths forward in action and then evaluating them as for as regards to which is the most beneficial path to take so already we're we're touching on the ground of active inference another form of influence is control and that comes in two flavors legitimate authority and coercion and legitimate authority we've already talked about this because we've talked about institutions and offloading of cognition from the individual level to the group level through social institutions of one kind or another norms or actual concrete institutions and things like that um and by and large legitimate authority people people are are are willing to uh prove to you know accept legitimate authority that because they understand that it's in their best interest you know one one example is we most of the work most of us

in the world are okay if uh if a traffic
cop stops someone who is
uh you know speeding at a very high
speed because that's dangerous to us all
and we
are glad the traffic cop is going to
take some action and exercise authority
over this fellow who's endangering the
rest of us
so by and large we we
we are good with legitimate authority
and we don't feel that it limits our
our freedom um and then there's coercion
which is more of a confrontational uh
form of influence where you maybe
force someone to to take us behave in a
certain way or to
um you know act in a certain way
and there's various forms of coercion
that can occur
uh including various forms of
manipulation i'll just list a few
but that could be disinformation deceit
emissions obstruction
the exercise of false authority drowning
out or silencing
of competing voices or various things so
you know this is rampant
in our societies today these these occur
you turn on the tv and you'll see
evidence of all of these
and uh turner makes the
point that coercion all of these forms
of coercion
is easily abused and very dangerous to a
society
because it re that tends to reduce
trust in societal institutions

it tends to reduce the long-term health
of the group
can be destructive in the sense that
coercion
breeds resentment to you know to being
forced to
behave in a certain way and it can
reduce
trust and reduce cooperation and it
increases an antagonistic climate
and uh that remains then that the
society who's during the coercion of the
group that's during the coercion has to
use even
more energy and more resources to
buckle down and make and make the
question even more severe so that they
can get
the individuals to behave in the way
they want
so uh you know there's a role for
coercion especially say like a you know
police uh uh
arresting a someone dangerous who has a
gun and
is about to shoot someone you would hope
that they would tackle that person to
the ground or whatever
and prevent them from doing that from
hurting others
so there's a role for coercion but it
can be easily abused as i said and must
be used
very transparently and judiciously or
else in the long run the whole society
is going to
suffer and potentially go down the tubes
um

um so that's the way he sees it and then
we can
we can see how this plays out in the
active active inference
framework already we've talked about
persuasion and the the similarities
between persuasion and learning
and then legitimate legitimate authority
as being similar to
offloading of cognition to insta social
institutions
and obviously when we're talking about
influence we're talking about group
associations
and we've already now mentioned that in
with respect to neuronal associations
and how the
how neurons and cis and people in a
society have to have
the freedom to associate at will to
to provide flexible cognition for the
society
um right so association is a part of
influence
just the fact that we're associating and
and and uh
you know trying to influence one another
is a part of cognition
so so what are the impacts of this on
design like how to how would we
how would how might we change a design
or or what kind of design
elements might we incorporate into new
societal systems if this was our concern
if if we wanted to encourage persuasion
and legitimate authority
how you know what this is the question
i'm asking

you know the world in this paper is
how how does how could this impact
design and you can think of a few things
off the top of your head
for example transparency would help this
process

uh coercion is very difficult to
achieve if there's high transparency
if there's quality information flowing
then coercion is more difficult to
achieve if there's quality education
happening maybe even especially
uh education about what coercion is what
what is manipulation what are the what
are the many faces of manipulation
how do different groups in society use
how do they how do they uh how do they
manipulate

in order in order to you know get their
will how is

uh how what is the role of media in
coercion you know these are all
these are all good questions to ask if
we're talking about

what kinds of new designs would we want
in order to avoid some of these problems
nice

the next of these four topics is
self-identity uh

and uh use the example of cancer here so
uh um levin has published a couple of
papers that i think are really
interesting on this topic

and uh it's it's cl it's true that as
cancer as a cancer develops in a body
the communication tends to up to be
changed

there there can be less quality

communication

between a cancer you know between a tumor and the surrounding tissues

and at the same time there can be more communication between cancer cells or within a tumor so

so communication kind of goes haywire in a cancer or can go haywire

and uh the

the idea that levin has is that what happens is

we often we often we often portray a cancer as being a selfish

you know a selfish growth that's occurring in the body

and he says you know it's not really quite like that it's uh

it's a little bit more like both normal tissue

and cancer tissue is self-interested and acting in a self-preserving manner the difference

is what the extent you know the functional extent of self

is for the two populations so the functional extent itself for

for normal tissue normal cells in a normal tissue is the whole body like the cells are connected and are communicating with

other cells in the body both locally and distantly so the

the body is a

cognitive apparatus you could say an extension of an individual cell

outward to a larger cognitive apparatus a larger environment a larger milieu

and in a cancer cell that those

communication that communication is
breaking down
and this the functional extent of cell
for a cancer cell might be
you know might grow smaller and smaller
and smaller so the cancer cell
just like the normal cell is acting in a
self-interested and self-preserving
manner
only its functional extensive self is
small
so uh it's taking care of us it's
including
in its self a small group of cells
small tissue as uh relative to the body
so it's it's its ability to to
think in a sense cognate shrinks because
it's no longer attached to the rest of
the cognitive apparatus of the body
it's has to do any cognitive processing
that it can do
within a much smaller set of cells so
it's capacity to act say over multiple
time scales to
behave in such a way that it will exist
in two years or five years is reduced
because it's it's he has a nice
term for it he calls it the cognitive
light cone of self
is reduced in the cancer and this is
really interesting you know when we talk
about us
how this might play out it's at a
societal level right
there's uh you know there's there's
maybe some injury or stress in a certain
population
that population is is retracting itself

from maybe an from the rest of the
society the rest of the environment
because it's maybe feels threatened or
it feels
unheard or it feels unappreciated or if
just is under stress
and uh it's you know the the
local group is forming that is more
interested in the local
cognitive uh light cone of self
a smaller like cone of self uh
and that can be dangerous you know if if
that happens too much in a society that
can be dangerous
it can also be useful because people are
getting together in small groups to
think about how to solve a problem that
can also be useful
but but if that is if there's too much
of that or if it happens under the wrong
circumstances
then you can get the situation where the
society is really kind of
starting to disintegrate because because
there are elements in the society that
are acting
preserving the small self and acting to
you know protect the small self as
opposed to the whole
and you know you can just imagine what
that leads to
and so the so the task of design
then is to create a a cognitive
framework
that where in which communication is
rich in which everyone's voice is heard
in which
problems are actually addressed and not

just

glossed over or ignored where where
the needs of uh you know the whole are
taken into account

uh where everyone's story matters
where everyone has an opportunity to
impact the

larger workings of the society the
decisions that the

society makes and uh you know
building such a design building such a
society

and a cognitive system then would reduce
the risk

of you could say cancerous growths
occurring

uh right so uh you know the cancer is
informative in this sense you know it's
an amazing i what my background is in
cancer biology

and it's a never-ending source of
information and

you know to look at how complex the
complexity of the
situation surrounding cancer

so that's it so you know we can design
systems so that cognitive light cone
itself

is expanded rather than contracted and
that's going to go a long ways towards
towards you know kind of encouraging
cooperation

encouraging coordination encouraging
group problem solving on group problems
you know nice thank you

the last topic is uh power distributions
so we've been talking about uh
functional systems each functional

system has a function i've listed some of those and each functional system also operates via a particular medium so for example you know the a good example is the economic system operating under with the through the medium of money uh so some authors that i cite suggest that each functional system if left unchecked tends to have a growth imperative and we certainly see that in the economic system it's like everything is now suddenly uh my couch and my living room is now worth money because i can sell it to someone who wants to visit my hometown for a weekend right so everything is commoditized now or many many things are commoditized that used to be more freely shared without just in the course of life with uh with people who had needs so i sell my i sell my time i sell my resources i sell my ideas and uh and and and financial the financial system seems to have overtaken many aspects of our life so in that sense in our current society you might say that the the financial the economic system is over emphasized relative to the other functional system say the education system and other systems so uh

obviously then when we're talking about design one of the goals of design is to balance the the influence of each function system and that is a topic that we already mentioned very briefly about in integrating systems together so integrated economic system legal system governance system you know all serving the same purpose and all acting as a as a flexible uh cognitive architecture for the greater good of the of the whole if i can give a comment on that uh you mentioned checks and balances and that in the american context referred to several separations within the governance mechanisms and the development that happened subsequent here we're talking about checks and balances in a little bit more of a general cybernetic goal-oriented system way and we're talking about checks and balances between these different sectors it almost sounds like so it's not just that there's going to be market mechanisms with all of the manufacturers of a product competing on that one dimension and then on a separate sector there's competition of the academics on this one question in this one way how can we link those competitions in a way that actually is aligned with the overall goals of

society right exactly
how can we align all of the systems to
the same purpose
which is the the you know the the
reduced uncertainty and expected
uncertainty of the
of the society in relation to the
essential variables the things that
matter
right so that's one uh aspect of design
on this
is how can we maintain a balance among
systems
but another aspect is how can we
maintain a balance within systems
so the economic system again provides a
really good example here
and i i say i have a sentence there that
today the wealthiest one percent of
individuals globally own about 40
percent of the world's wealth
now that is like insane i mean that is
crazy especially if we think of money as
a voting tool
i mean money is a voting tool it's it's
an info it's
not a recognized formal voting tool
right
but it's a undemocratic one if i have if
i
if i'm a billionaire i have lots of
votes in this world i can get people to
do
any number of things that i wish them to
do i can direct
non-profits to focus on particular
issues i can
i can make a new company that does a

particular

you know provides a particular service

or or

you know i can i i can i have all

kind i have a billion times more

influence than

just a normal person right so money is a

voting tool

and it's just uh it's just used as an

undemocratic one

but it could be used as a democratic

voting tool it could

the money system the economic system

could be

highly transparent it could

it could achieve a um a

reasonable degree or even a strong

degree of income and wealth

equality so that the money as a voting

tool becomes a fair voting tool

right like in a democracy we would not

we would not stand for

i get one vote and some and the guy down

the street gets a billion votes i mean

that would just we would recognize that

as being deeply unfair

but it's the same thing in an economic

system if if one person has one dollar

another and the next guy has a billion

dollars

that person has enormous control over

how

uh society makes decisions so it's not

just economic power

i mean i'm focusing on the economic

system but there's you know there's six

or ten systems that we could be focusing

on here and each of them has some kind

of medium

and we could talk about the distribution

of power in any of them

for example we could talk about the

distribution of power in a

in a governance system currently in the

u.s

there are what 400 i don't know how many

how many legislators and senators do we

have in total i don't know what that

amounts to but

compared to the population this is a

time we have a we have empowered

we have put enormous power in the hands

of a tiny sliver of

people and you know does is that a good

idea

in today's world could we do better i

imagine

that a different kind of governance

system would destroy would do a far

better job of distributing power

so that everyone can have feel like

their voice is heard and that they can

be part of the decision-making process

you know as it stands right like right

now in america i get i get to vote every

two years

uh about like a yes i get like a thumbs

up or a thumbs down

on certain candidates and that's that's

kind of it formally that's it i mean i

can do some informal things i can

write to my congressman or i can you

know i can

complain in the newspaper or write a

letter to the editor or something but as

far as formal

things that i can do to have my very
nuanced

understanding of the world heard all i
all i really get is a is like a yes or
no every few years you know and then
a group of candidates and that's hardly
that's hardly taken advantage of my
understanding of the world right i i
have much more

uh insight to offer than just a simple
yes or no every few years
so i think it would be possible then to
uh

to to design systems that balance
this medium of communication
in each of the function systems achieve
some much greater degree of incoming
wealth equality
achieves a much greater degree of power
distribution within governance systems
uh power distribution within legal
systems

uh you know there's a great room for
improvement
on all of these

uh yeah so i think is there one more oh
there's one more slide

yep and uh that is really focused on
economic

power so the first one i already talked
about using money as a
voting tool we already do that and we
would be better if we just

did that above board and in a democratic
and transparent fashion

and new systems could do that and when i
say economic systems again i am
including in that monetary systems

financial systems and the bigger picture
um a large set of studies have shown
that incoming wealth inequality is bad
for society
it affects uh health cooperation trust
anxiety happiness
social cohesion and then mortality and
uh social stability in a number and
numerous other factors so
it's quite clear that high amounts of uh
any income and wealth inequality are
detrimental to society
and and again today we are extremely
uh unequal extremely unequal
and then uh the last just to end this
whole uh
talk today i would like to point out
that income equality
is really not a there's a it's kind of
it's not unpopular the concept is not
unpopular
there was a large study done a few years
ago where they
showed they didn't they didn't they
showed
participants uh graphs of the division
of economic power
income and things like that uh two you
know they had three different choices
an image that was really uh very equal
like
50 50 equal one that was uh fairly equal
but not quite you know not quite
perfectly equal
and then a really extreme one well it
turned out that
people didn't you know they asked you
ask people like which of these do you

prefer

and it turned out that they didn't like

the extreme one at all

and that happened to be the one that was

the picture of the u.s

inequality in the u.s and uh and the one

that was more mild i think was the

inequality maybe in

in sweden or something like that so they

they preferred the the distributions of

economic power that were less

less extreme and when asked

which of these do you think is the most

fair for the society

the the the uh more than half of those

queried and this was many thousands of

people who were queried

chose actual income equality like you

know like

completely in equal incomes as the most

fair way

to run a society and i don't i don't

suppose that they

that people really thought that was

possible like you know that's i know

it's not possible but that would be the

more fair way that's what i imagine

people were thinking

and uh when we next come back to talk

we'll talk about the leader framework

and some

potential designs that actually would

achieve

income equality income and wealth

equality over time

so that's that's yet to come

wow awesome that really flew by that i

uh after some little tech bumps

it was awesome and that is a little bit
like an analogy for maybe
this program there's some technical
details up front
few live streams to start a few few
teams few memes
but um maybe once there's momentum
and once there's proof in the world and
a little bit more of
a contact between the theory and the
practice we would see
some new kinds of momentum so maybe give
a last
comment what is going to happen in the
fourth session and then our two
group discussions and um yeah yeah we're
gonna work we have one more uh
i'm glad you said that because i did
want to say to folks that we have one
more discussion
in this series coming up next week and
in that discussion we'll talk about the
leader framework as a prototype and some
of the things it might achieve how it
might work
and that
is a big part of our discussion next
time
and then there's two regularly scheduled
regular scheduled discussions on part
three of the paper that will
happen in may
i think the first one is in two weeks
it's the first two
tuesdays of may at 7 00 a.m pacific
and that will be like the group active
inference live stream so we'll have a
bunch of other

colleagues and that will be more of a
group discussion but the fourth
conversation between just us will be
april 27th

at 3 30 pm pacific so i'm really looking
forward to that

and thanks for um for coming on for so
many times and giving us so many
good insights and we appreciate your
time and all the participants who stuck
with us also for the tech

thanks thanks for having me after